

Introducing Linear Equations

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Lesson Plan: Introducing Linear Equations

Year Level: Year 7

Learning Area: Mathematics

Duration: 60 minutes

Topic: Linear equations

Curriculum grounding: VCAA Mathematics Version 2.0, Level 7 Algebra. This lesson aligns with the expectation that students solve one-variable linear equations with natural number solutions and verify solutions by substitution.

Learning Intention

Students will understand that a linear equation is a mathematical statement containing an unknown value, and that solving an equation means finding the value that makes the statement true.

Success Criteria

Students can:

- identify the unknown value in a simple linear equation
- explain why the equals sign means both sides have the same value
- solve one-step equations using inverse operations
- check a solution by substituting it into the original equation

Materials

- Whiteboard or projector
- Mini-whiteboards
- Student notebooks
- Balance-scale image or diagram
- Practice questions

Lesson Sequence

1. Hook: What keeps an equation balanced? (10 minutes)

Write this on the board:

$$\square + 5 = 12$$

Ask students what number belongs in the box and how they know. Discuss the idea that both sides of the equals sign must have the same value. Show a simple balance-scale diagram and connect it to the equation.

Teacher questions:

- What does the equals sign tell us?
- How could we prove the missing number is correct?
- What would happen if we changed only one side?

2. Explicit teaching: From boxes to variables (10 minutes)

Explain that mathematicians often use letters instead of boxes to represent unknown values.

Replace the box with x :

$$x + 5 = 12$$

Define a linear equation as an equation involving a variable where the variable is not raised to a power. Explain that solving means finding the value of the variable that makes the equation true.

Model:

$$x + 5 = 12$$

$$x = 7$$

Then check:

$$7 + 5 = 12$$

3. Worked examples: Using inverse operations (15 minutes)

Model three examples with clear thinking aloud.

Example 1:

$$x + 6 = 14$$

Subtract 6 from both sides.

$$x = 8$$

Example 2:

$$x - 4 = 9$$

Add 4 to both sides.

$$x = 13$$

Example 3:

$$3x = 18$$

Divide both sides by 3.

$$x = 6$$

After each example, substitute the solution back into the original equation.

4. Guided practice (10 minutes)

Students solve questions on mini-whiteboards while the teacher checks responses.

Questions:

- $x + 4 = 11$
- $x - 3 = 8$
- $5x = 25$
- $x + 9 = 17$

Pause after each question to discuss the inverse operation used and how students can check their answer.

5. Independent practice (10 minutes)

Students complete a short practice set.

Core:

1. $x + 7 = 15$
2. $x - 6 = 10$
3. $4x = 20$
4. $x + 8 = 19$

Extension:

1. $2x + 3 = 11$
2. $3x - 4 = 14$

6. Plenary and exit ticket (5 minutes)

Students answer:

1. What does it mean to solve an equation?
2. Solve $x + 9 = 16$.
3. How can you check that your solution is correct?

Differentiation

- Support: Use number boxes before variables, balance diagrams, and sentence stems such as "I undo *by*."
- Extension: Introduce simple two-step equations and ask students to explain each inverse operation.

Assessment

The teacher uses mini-whiteboard responses, questioning during guided practice, and the exit ticket to check whether students can solve one-step linear equations and verify solutions by substitution.